

Technical Section

Network power diagrams with maximum-capacity constraints for embedded network partitioning[☆]

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ABSTRACT

Power diagrams are fundamental weighted partitioning structures in computational geometry and computer graphics. Existing studies have extended Voronoi diagrams from Euclidean domains to networks by replacing straight-line distances with shortest-path metrics. However, network counterparts of power diagrams that preserve connected power cells and satisfy maximum-capacity constraints remain insufficiently explored. This paper presents maximum-capacity constrained network power diagrams (MCCNPD) for embedded network partitioning. We first define a network power diagram using a squared shortest-path power distance on an embedded network graph. To generate connected network power cells, we construct them through a rooted connectivity-preserving growth process. Based on this, maximum-capacity constraints are imposed on network power cells, and a two-stage optimization strategy is developed to compute capacity-feasible partitions. The first stage performs weight-driven global deformation of network power cells. The second stage applies frozen-weight local refinement to resolve residual capacity violations while preserving cell connectivity. The formulation is further extended from discrete demands to continuous embedded network demand densities via line-integral capacity accounting. Experiments on representative synthetic networks and a real urban road network show that the proposed method produces connected network power partitions and effectively reduces capacity violations under both discrete and continuous demand models.

1. Introduction

Geometric region partitioning [1] is a fundamental operation in computational geometry, computer graphics, and spatial analysis. Classical Voronoi and power diagrams [2] provide compact structures for decomposing a domain, and they have been widely used in geometric processing [3], mesh generation [4], and spatial optimization [1]. Within these fields, the embedded network partitioning constitutes an essential application of geometric partitioning [5], which has practical significance for urban spatial planning and road network management.

Existing studies provide important geometric foundations for embedded network partitioning. The network Voronoi diagram (NVD) [6] extends classical Voronoi tessellations [7] to graphs by replacing Euclidean distance with network shortest-path distance, while weighted NVD (WNVD) [8] further incorporates site or network attributes into the distance model. They are effective for proximity-based region partitioning on networks, where each network region is mainly determined by network shortest-path proximity. However, they do not provide site-specific weights as optimization variables for explicitly deforming

network cell influence. Moreover, a direct extension from network shortest-path assignment to squared network shortest-path power distance may produce fragmented and disconnected regions on graphs, leading to topological inconsistency.

Capacity constraints introduce an additional requirement for embedded network partitioning [9]. In many practical partitioning scenarios, each region is expected to satisfy a prescribed value or an upper capacity bound. The capacity-constrained Voronoi and power diagrams [10] have been investigated mainly in Euclidean domains, where site weights can be adjusted to control cell area or mass. More recent studies have introduced the capacity constrained network Voronoi diagram (CCNVD) [11] and regionalization approaches [12] for generating contiguous or balanced regions on graphs. However, they are usually based on proximity assignment or combinatorial graph partitioning [13] rather than network power-distance geometry. As a result, existing methods either retain weighted geometric interpretability without network topology, or enforce network contiguity and capacity without a power-diagram structure on network graphs. A unified formulation that simultaneously supports shortest network-path geometry,

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site weights, connected cells, and maximum-capacity constraints on embedded networks remains insufficiently explored.

To address this, we propose the maximum-capacity constrained network power diagram (MCCNPD) for embedded network partitioning. We first define network power distance using squared network shortest-path distance and site-specific weights on an embedded network graph. Then, we construct connected network power diagram (NPD) through a rooted connectivity-preserving growth process. The maximum-capacity constraints are further imposed through a two-stage optimization strategy that combines weight-driven global deformation with frozen-weight local refinement. Moreover, the proposed framework is extended from discrete demand elements to continuous network demand densities through line-integral capacity accounting. The main contributions are summarized as follows:

- We introduce a network power diagram formulation on embedded networks based on squared shortest-path network power distance.
- We propose a connected-by-construction NPD algorithm that generates rooted connected network power cells and supports segment-level boundary splitting.
- We develop a two-stage MCCNPD optimization framework that enforces maximum-capacity constraints and extends to continuous network-density fields via line-integral capacity accounting.

The remainder of this paper is organized as follows. Section 2 reviews related work on capacity constrained network Voronoi and power diagrams, network Voronoi diagrams, and capacity constrained network partitioning. Section 3 presents the discrete MCCNPD formulation, connected construction, and two-stage optimization strategy. Section 4 introduces the continuous-density extension. Section 5 reports experiments on representative synthetic embedded network graphs. Section 6 demonstrates a real-world application study, and some conclusions are provided in Section 7.

2. Related work

In this section, we briefly review related work on Voronoi and power diagrams, network Voronoi diagrams, and capacity constrained network generation.

2.1. Voronoi and power diagrams

Voronoi diagrams [7] are fundamental structures in computational geometry. Given a set of sites, they partition a domain into cells according to nearest-site proximity. Their geometric definition and combinatorial properties have made them useful in geometric modeling, mesh generation [3], spatial optimization [14], and graphics applications [15]. Aurenhammer [2] systematically reviewed the theoretical foundations and construction algorithms of Voronoi diagrams. Recent studies continue to extend Voronoi computation to more complex geometric applications, including triangular meshes [4], differentiable simulation models [7], and layout generation [16]. However, the nearest-site proximity in Voronoi diagrams indicates that Voronoi cells are determined with given site positions.

Power diagrams [17] further extend Voronoi diagrams by assigning a weight to each site. A power cell is defined by a weighted distance function, called power distance, which provides direct control over cell size and mass distribution. This property makes power diagrams particularly important in prescribed-capacity tessellation [18], centroidal power diagram [10], medial-axis computation [17], and geometry processing [1]. Recent work has further developed restricted power diagrams [15], meshless power diagrams [19], generalized balanced power diagrams [20], and differentiable Voronoi or power-type structures [7] for more flexible geometric computation. Compared to Voronoi diagrams, weights provide a principal mechanism for deforming cells while retaining a clear geometric interpretation.

However, existing Voronoi and power diagram formulations are still predominantly designed for Euclidean, volumetric, or surface domains. Their cells are induced by continuous geometric distance functions, whereas network cells are connected subgraphs induced by shortest-path distances. Therefore, these methods provide the geometric foundation for NPD, but they do not address power diagram construction on embedded network graphs.

2.2. Network voronoi diagrams

Network Voronoi diagrams extend Voronoi tessellations from Euclidean space to network space, where the domain is represented as a graph and distances are measured by the shortest paths along network edges. The generalized NVD [6] provides a systematic framework for defining proximity cells on networks with different generator types and distance models. Compared with planar Voronoi diagrams, the NVD better represents accessibility and proximity on networks under network-constrained movement. Subsequent studies [21] have applied Voronoi-based network partitioning to network matching, traffic analysis, accessibility modeling, and multi-modal transport networks [22].

The WNVD further incorporates site attributes, edge attributes, or event distributions into network-based partitions [23]. She et al. [8] have applied WNVD for local spatial analysis by deriving network weights from events constrained to street networks. Other graph-based Voronoi variants have also been developed for more complex network settings. Transit Voronoi diagrams extend network Voronoi ideas to public transport systems by considering transfers, schedules, and changing network topology [24]. More recently, graph Voronoi partitioning has been used on dynamic topological graphs for multi-UAV exploration, where graph nodes are allocated according to motion cost rather than Euclidean distance [25]. These studies confirm that Voronoi-based partitioning can be generalized from Euclidean domains to graph-structured network domains.

Nevertheless, existing network Voronoi and graph-based Voronoi methods remain primarily proximity-driven. Their cells are usually determined by the shortest-path or motion-cost assignment, possibly modified by additive, multiplicative, or attribute-dependent weights. They do not define site-specific power weights as optimization variables for explicitly deforming network cell influence. Moreover, direct assignment under squared shortest-path power distance may produce separated cell fragments on complex embedded graphs. In contrast, the proposed NPD formulates network power distance on network graphs and combines it with connected cell construction.

2.3. Capacity constrained network partitioning

Capacity constraints are widely investigated in spatial partitioning, districting, facility assignment, and network service-region design. Capacity constrained Voronoi and power diagrams [10] in Euclidean domains can be interpreted through weight adjustment. That is, the weights deform cell to satisfy prescribed area or mass constraints. The CCNVD [11] in network domains provides a representative approach for generating contiguous graph regions under capacity requirements. Related studies on embedded network partitioning [9], political districting [26], p-median models [27], and location-allocation optimization [28] also consider contiguity, balance, demand assignment, and capacity feasibility. These works provide useful optimization models for capacity-aware network partitioning.

Recent constrained geometric partitioning studies further demonstrate the usefulness of Voronoi-type structures in allocation and resource distribution problems. Generalized balanced power diagrams [20] provide a flexible model for balanced tessellations. Relaxation-based Voronoi approaches have been used for equitable resource distribution under demand coverage and equity constraints [29]. Voronoi-based task allocation, multi-agent coverage, and robot delivery methods also introduce practical constraints [30] such as workload balance, robot capacity, non-convex domains, and relay-network structure.

However, capacity-aware network partitioning and power diagram construction have largely been developed. Capacity constrained power diagrams retain weighted geometric interpretability but are mainly defined in Euclidean domains [31]. Network partitioning and regionalization methods [5] can enforce contiguity and capacity constraints, but they are usually formulated as assignment or graph-partitioning models rather than power-distance-induced geometric diagrams. As a result, existing methods either preserve power diagram geometry without network topology, or preserve network feasibility without a power diagram structure. The proposed MCCNPD framework addresses this by constructing connected power cells and imposing maximum-capacity constraints on embedded network graphs.

3. MCCNPD with discrete demands

In this section, we present the discrete formulation of network power diagrams with maximum-capacity constraints, the connected-by-construction NPD method, and the two-stage MCCNPD optimization strategy.

3.1. Problem definition

Let $G = (V, E)$ be an embedded network represented as an embedded graph, where V is the set of network vertices and E is the set of edges representing segments. Each edge has a positive length, and the distance between two points on the network is measured by the shortest-path distance $d_G(\cdot, \cdot)$. Let $S = \{s_i\}_{i=1}^m$ be a set of sites on the network, and let $W = \{w_i\}_{i=1}^m$ denote the associated weights. For any point $x \in G$, the network power distance from x to site s_i is defined as

$$d_G^p(x, s_i) = d_G(x, s_i)^2 - w_i. \quad (1)$$

Following the terminology of power diagrams, we refer to this squared shortest-path weighted function as the network power distance. Given S and W , the network power diagram is then denoted by $P(S, W) = \{P(s_i, w_i)\}_{i=1}^m$, where each network power cell $P(s_i, w_i)$ is associated with site s_i . A network point x belongs to $P(s_i, w_i)$ if s_i gives the smallest network power distance among all sites, namely

$$P(s_i, w_i) = \{x \in G \mid d_G^p(x, s_i) \leq d_G^p(x, s_k), \forall k \neq i\}. \quad (2)$$

The notion $P(S, W)$ is used for the whole diagram, while $P(s_i, w_i)$ denotes an individual network power cell.

Let $Q = \{(q_j, \mu_j)\}_{j=1}^n$ be a set of demand points on the network, where q_j is the demand location and $\mu_j \geq 0$ is the demand value. The assigned demand, or cell mass, of $P(s_i, w_i)$ is the total assigned demand:

$$m_i = \sum_{q_j \in P(s_i, w_i)} \mu_j. \quad (3)$$

Each site s_i is associated with a maximum capacity $c_i^{\max}$, and $\sum_{i=1}^m c_i^{\max} \geq \sum_{j=1}^n \mu_j$. A capacity-feasible cell must satisfy $m_i \leq c_i^{\max}$. Therefore, the discrete MCCNPD problem is to construct an NPD on G such that each network power cell remains connected on the graph and satisfies the maximum-capacity constraints.

3.2. Connected NPD construction

Given the sites S and the related weights W , a direct NPD construction assigns each network point to the site with the minimum network power distance. However, this pointwise rule may assign disconnected graph components to the same site on an embedded graph, as shown in Fig. 1(a-b). To avoid this topological inconsistency, we propose a connected-by-construction NPD method through a rooted graph-growth process, as shown in Fig. 1(c-d). The pseudocode is provided in Algorithm 1.

Specifically, each site located in the interior of a segment is first inserted into the graph as a root vertex. For each vertex $v \in V$, we

Algorithm 1 Connected NPD construction strategy

Require: Network graph $G = (V, E)$, sites S , weights W

Ensure: Connected NPD $P(S, W)$

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1: Insert site positions into  $G$  as root vertices
2: Compute shortest-path distances from sites to vertices
3: Initialize each network power cell  $P(s_i, w_i)$  with site  $s_i$ 
4: Insert neighboring vertices into a priority queue
5: while unassigned vertices exist do
6:   Extract the frontier candidate with the minimum network power distance
7:   if  $v$  is unassigned and adjacent to  $P(s_i, w_i)$  then
8:     Assign  $v$  to  $P(s_i, w_i)$ 
9:     Insert unassigned neighbors of  $v$  as new candidates
10:  end if
11: end while
12: for each edge  $e = (u, v) \in E$  do
13:   if endpoints  $u$  and  $v$  have the same owner then
14:     Assign the edge  $e$  to the related network power cell
15:   else
16:     Solve Eq. (5) and split the edge at boundary point
17:   end if
18: end for
19: Return the connected network power diagram

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compute the shortest-path distances from all sites to all graph vertices according to the network power distance in Eq. (1). Based on this, the construction starts from all site roots simultaneously, and each site root is initialized with a network power cell, as shown in Fig. 1(c). A priority queue stores frontier candidates keyed by their network power distances. Each candidate is represented by a site-vertex pair (s_i, v) , where v is unassigned and adjacent to the current network power cell $P(s_i, w_i)$. At each step, the candidate with the smallest network power distance is extracted. It is assigned to $P(s_i, w_i)$ only if the vertex is still unassigned and remains adjacent to the current network power cell $P(s_i, w_i)$. The unassigned neighbors of the newly inserted vertex are then added to the queue. This procedure guarantees rooted connectivity because each newly assigned vertex is attached to an already assigned vertex in the same network power cell. Therefore, each vertex-level cell remains connected to the site root during the entire construction process.

After vertex ownership is obtained, each segment is further processed to generate segment-level network power cells. For an edge $e = (u, v) \in E$ with length l_e , the whole segment is assigned to their common owner if u and v have the same owner. Otherwise, the segment is treated as a boundary segment. Let the endpoint owners be s_i and s_k . For a point $x(t)$ on the edge, where $t \in [0, l_e]$ is measured from u , we use an endpoint-owner-consistent split. The two side-consistent power functions are defined as

$$\begin{cases} \psi_i(t) = (d_G(u, s_i) + t)^2 - w_i, \\ \psi_k(t) = (d_G(v, s_k) + l_e - t)^2 - w_k. \end{cases} \quad (4)$$

The boundary position t^* is obtained by solving the network-power-distance equality, that is, $\psi_i(t^*) = \psi_k(t^*)$. Based on the squared network power distance, the boundary position is given as:

$$t^* = \frac{(d_G(v, s_k) + l_e)^2 - d_G(u, s_i)^2 + w_i - w_k}{2(d_G(u, s_i) + d_G(v, s_k) + l_e)}. \quad (5)$$

The subsegment $[0, t^*]$ is assigned to the cell of u , and the subsegment $[t^*, l_e]$ is assigned to the cell of v . This split is a side-consistent boundary split conditioned on the endpoint owners produced by the connected graph-growth process. This side-consistent formulation avoids ambiguous multiple intersections that may arise when the full piecewise shortest-path distance is evaluated along an edge while preserving the connectivity of the constructed cells.

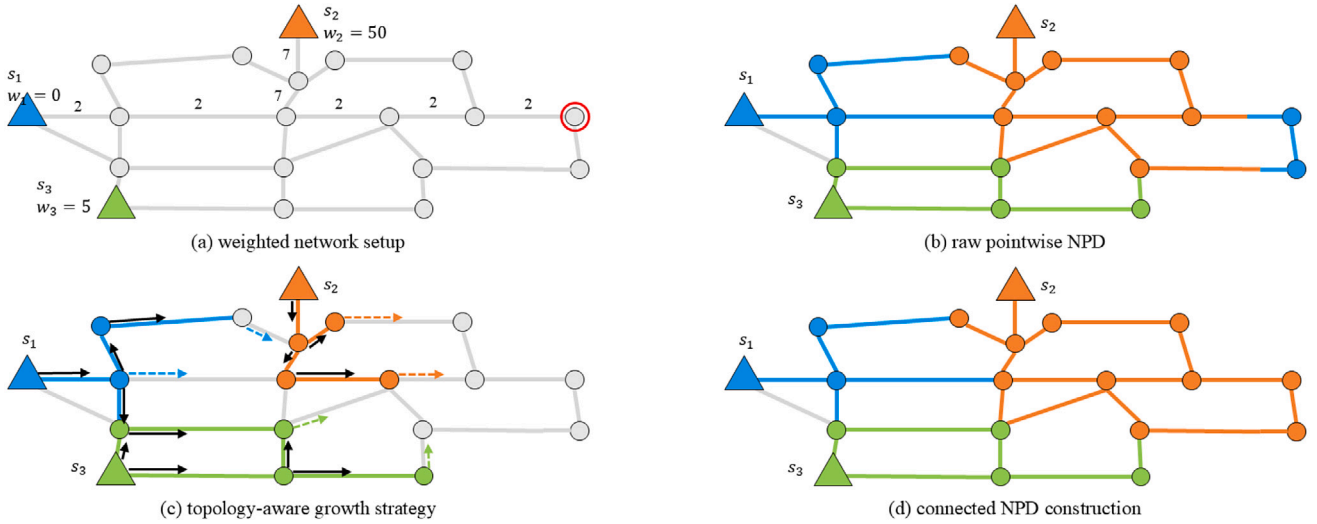


Fig. 1. Connected network power diagram construction on an embedded graph. (a) A weighted network with three sites, where direct evaluation of the network power distance along a reference route induces an alternating ownership sequence. (b) Direct pointwise network power distance may assign separated graph components to the same site, producing disconnected network power cells. (c) The proposed construction grows cells from site roots through network-power-distance-guided frontier expansion. Each newly assigned vertex is attached to an existing vertex in the same cell. (d) The resulting network power diagram preserves rooted cell connectivity. Boundary edges between different endpoint owners are split according to network-power-distance equality.

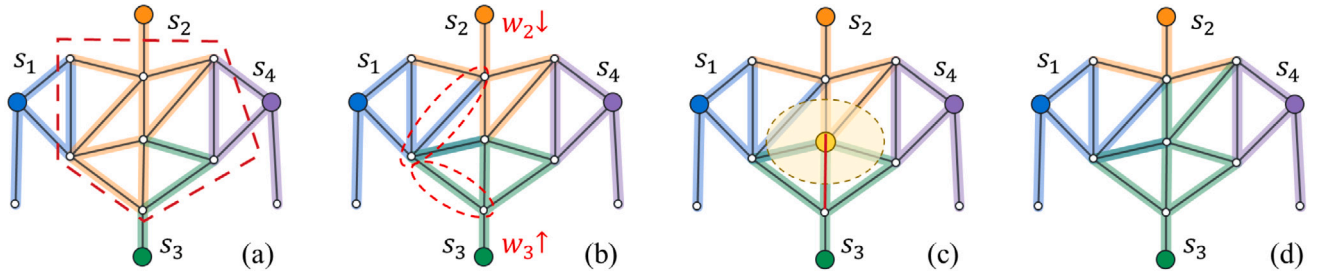


Fig. 2. Two-stage optimization for discrete MCCNPD construction, where all panels use the same embedded network topology, sites, and discrete demands. (a) The initial connected NPD provides a complete and connected partition, but may violate the maximum-capacity constraints. (b) The weight-driven global deformation stage updates site weights and reconstructs connected NPDs to reduce capacity violations. (c) The frozen-weight local refinement stage adjusts ownership within boundary corridors between overloaded cells and neighboring cells with remaining capacity. All accepted transfers preserve rooted cell connectivity. (d) The final MCCNPD assigns the embedded network to connected network power cells with maximum-capacity constraints.

3.3. Discrete MCCNPD optimization

The connected NPD construction provides topologically valid network power cells for fixed sites and weights. The maximum-capacity constraints are then imposed on these generated network power cells, that is, the assigned demand m_i of each network power cell $P(s_i, w_i)$ should not exceed the prescribed upper bound c_i^{max} . To achieve this, we develop a two-stage optimization strategy combining the weight-driven global deformation and frozen-weight local refinement, as shown in Fig. 2.

For each network power cell $P(s_i, w_i)$, we define the relative capacity residual as

$$g_i = \frac{m_i - c_i^{max}}{c_i^{max}}. \quad (6)$$

A positive value of g_i indicates that $P(s_i, w_i)$ violates the maximum-capacity constraint. The total capacity violation and maximum capacity violation are computed from the positive residual masses and are used to compare candidate partitions in the following stages.

Weight-driven global deformation. From the perspective of the network power distance definition, the modification of site weights directly changes the competitiveness of network power cells. Decreasing the weight of an overloaded cell weakens its influence, whereas increasing the weight of a neighboring under-capacity cell encourages

nearby demand to move into that cell after the connected NPD is reconstructed. Therefore, the site weights are iteratively updated to deform the connected network power cells in the first stage, as shown in Fig. 2(a-b).

At each iteration, we estimate a local power-margin scale for each cell at iteration t . For a demand point $q_j \in P(s_i, w_i)$, the margin to the best alternative site is

$$m_{ij} = \min_{k \neq i} [d_G(q_j, s_k)^2 - w_k] - [d_G(q_j, s_i)^2 - w_i]. \quad (7)$$

A small m_{ij} indicates that the ownership of q_j is sensitive to a small weight perturbation. For each cell, we collect the nonnegative margins into a set $M_i = \{m_{ij} | q_j \in P(s_i, w_i), m_{ij} > 0\}$. The values in M_i are sorted in ascending order, and $Q_\tau(M_i)$ denotes the empirical τ -quantile of this ordered sequence. Specifically, the τ -quantile is obtained at the position $1 + \tau(|M_i| - 1)$. If this position falls between two adjacent sorted values, linear interpolation is used, and the local scale is then defined as

$$S_i = \max(Q_\tau(M_i), \delta_i^2), \quad (8)$$

where δ_i is the shortest graph distance from s_i to its nearest neighboring site. The term δ_i^2 provides a geometric fallback scale when the margin set is sparse or unstable.

The update magnitude is determined by the local power-margin scale and the current capacity residual

$$\alpha_i = \max(\alpha_{min}, \eta S_i (1 + |g_i|)), \quad (9)$$

where η is a global adjustment ratio and α_{min} prevents vanishing updates. Let $N(i)$ denote the set of cells adjacent to $P(s_i, w_i)$, and $N^+(i) = \{r \in N(i) | g_r > 0\}$ represents the overloaded neighboring cells of $P(s_i, w_i)$. The weight variation Δw_i is computed as

$$\Delta w_i = \begin{cases} -\alpha_i, & g_i > 0 \\ \beta \alpha_i \min \left(1, \frac{c_i^{max} - m_i}{c_i^{max}} \right), & g_i \leq 0 \text{ and } N^+(i) \neq \emptyset \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

where β controls the extent to which under-capacity cells support neighboring overloaded cells. Thus, network power cells with large capacity residuals or stiff boundaries receive larger updates, while small-margin sensitive boundaries are updated more conservatively.

Frozen-weight local refinement. Due to the discrete and graph-constrained nature of the partitioning, the weight-driven stage may leave residual capacity violations near cell boundaries. Therefore, the second stage further applies the frozen-weight local refinement to eliminate residual capacity violations, as shown in Fig. 2(c-d). Specifically, we first identify overloaded cells as donors and adjacent under-capacity cells as receivers. For each donor–receiver relation, a boundary corridor is extracted around their shared interface, and the candidate refinements are subsequently evaluated within this local corridor.

The local refinement includes two types of reassignment operations. The first type is a single-demand reassignment. A boundary demand point q_j is directly transferred from a donor cell $P(s_a, w_a)$ to a receiver cell $P(s_b, w_b)$ if the receiver has enough remaining capacity, i.e., $m_b + \mu_j \leq c_b^{max}$. This operation is used to resolve small residual capacity violations by locally shifting individual boundary demands. The second type is a connected-block reassignment. Instead of moving one demand point, several neighboring demand points are grouped as a connected demand block B and transferred together. To construct such a block, we start from a donor-owned boundary demand adjacent to the receiver cell and expand within the donor cell along network adjacency. Neighboring donor-owned demands are gradually included until the block reaches a prescribed maximum size, or provides sufficient transferred demand for the local capacity repair. Thus, the block represents a compact group of adjacent demands near the donor–receiver boundary. A connected demand block B can be transferred from the donor to the receiver if the receiver capacity remains feasible, i.e., $m_b + \sum_{q_j \in B} \mu_j \leq c_b^{max}$. This block reassignment is useful when a single demand transfer is insufficient or when several adjacent demand points need to be moved together to maintain graph connectivity. Consequently, the optimization terminates when all network power cells satisfy their maximum-capacity constraints or when no feasible local refinement remains. The final partition preserves connected network power cells and satisfies the prescribed capacity bounds if a feasible refinement is found.

3.4. Algorithmic analysis

The MCCNPD optimization is designed as a feasibility-oriented heuristic. Specifically, a small weight change may induce a discrete change in the network power cell construction. The proposed optimization algorithm therefore uses capacity violation as the guiding measure and combines global weight-driven deformation with local connectedness-preserving refinement. For a network partition, the assigned demand of network power cell $P(s_i, w_i)$ is denoted by m_i , and its maximum capacity is c_i^{max} . The capacity violation of $P(s_i, w_i)$ is $v_i = \max(0, m_i - c_i^{max})$, and the total violation is $V = \sum_i v_i$. Based on this, we monitor the number of violating network power cells and the maximum single-cell violation. These quantities form the acceptance criteria of the heuristic search. A candidate partition is preferred when it reduces the total violation, reduces the number of violating cells, or

decreases the maximum violation without breaking network power cell connectedness.

In the weight-driven global deformation stage, the role of the weights is to deform the network power distance field. Similar to the classical power diagram in continuous spaces, for a cell with excess demand, decreasing the weight increases the effective power distance and makes it less competitive near the boundary. For an adjacent under-capacity cell, increasing the weight decreases the effective power distance and allows it to absorb nearby demand. Thus, the first optimization stage utilizes this mechanism to perform a global deformation of connected network power cells. However, the network power cell changes can be discontinuous, the first stage keeps the best incumbent partition according to the violation measure, but this may leave residual capacity violations near cell boundaries.

The frozen-weight local refinement stage then further refines the incumbent partition with fixed weights. A local transfer is accepted only if it preserves the connectedness of the affected cells and strictly improves the capacity-violation state. Since the number of discrete assignments is finite, a sequence of strictly improving accepted transfers cannot revisit a previous assignment. Thus, the local refinement cannot cycle and must terminate. The algorithm stops when all cells satisfy the maximum-capacity constraint or when no admissible improving transfer remains.

4. Continuous-density extension

The discrete MCCNPD represents demands as a finite set of points on the embedded network. However, in some cases, demand is more naturally modeled as a density distributed along network segments. In this section, we extend the proposed method to continuous network demand densities.

4.1. Network density representation

To support a continuous mass distribution on the embedded network, demands are represented as a nonnegative density field $\rho(s) \geq 0$ directly on the embedded graph G . The continuous-density representation follows the strategy of weighted network Voronoi analysis, where discrete demands are first projected onto the embedded network and then converted into a density field evaluated on network elements [8]. In this paper, the embedded network is subdivided into micro-segments for numerical integration. Rather than being tuned for each case manually, the target micro-segment length is determined from the total network length as

$$h = \max(h_{min}, L_G / N_{micro}), \quad (11)$$

where L_G is the total embedded network length. Therefore, each segment is subdivided into short micro-segments, and the density value is evaluated at the midpoint of each micro-segment. The density field is then estimated and normalized over the micro-segments.

For a network power cell $P(s_i, w_i)$, the assigned demand m_i is computed by line integration over the corresponding cell $m_i = \int_{P(s_i, w_i)} \rho(s) ds$. Using the micro-segment representation and considering maximum-capacity constraints, this integral is approximated as:

$$m_i \approx \sum_{c \in P(s_i)} \rho_c \ell_c \leq c_i^{max}. \quad (12)$$

4.2. Continuous-density MCCNPD optimization

The continuous-density optimization follows the same two-stage strategy as the discrete MCCNPD solver. In the weight-driven global

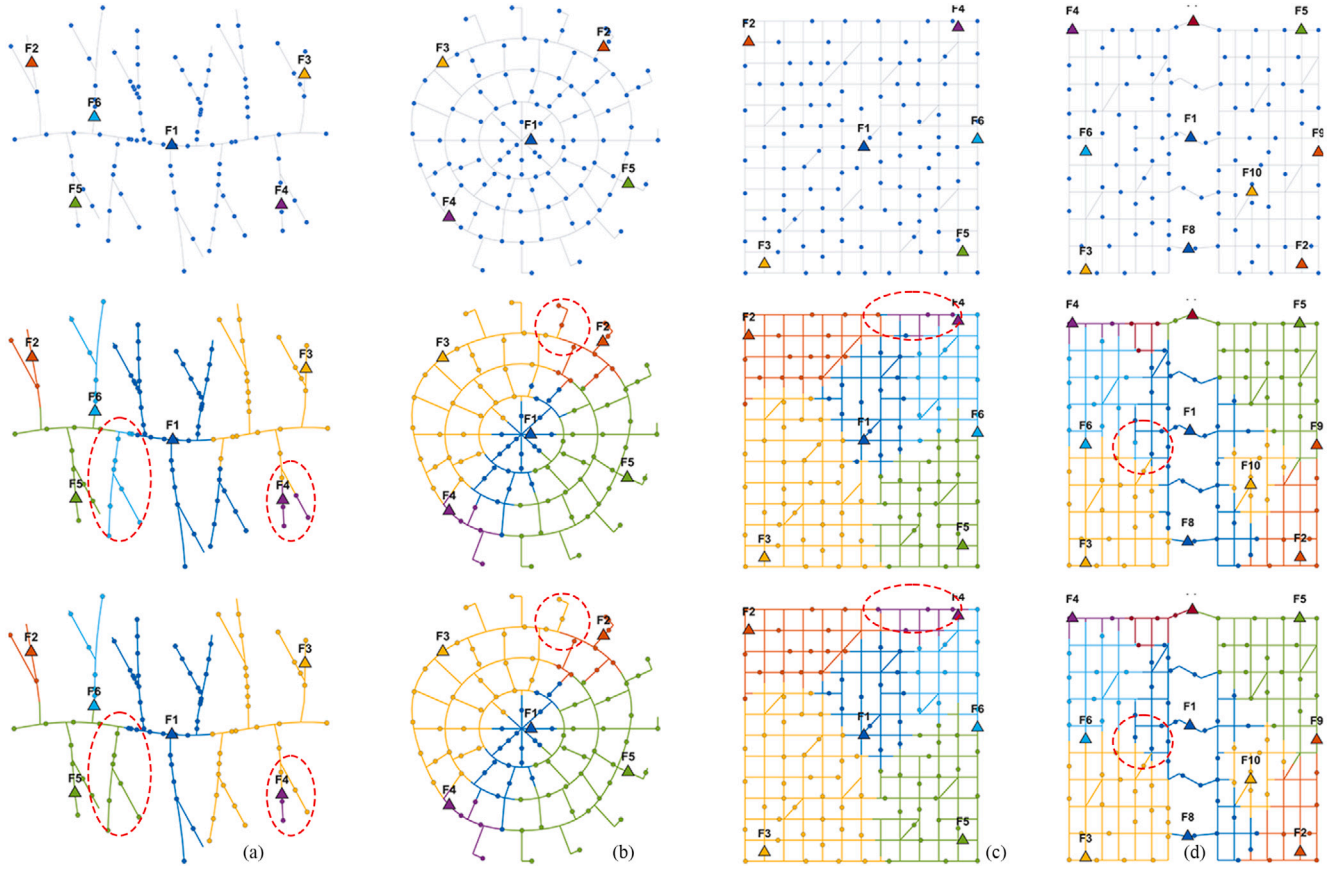


Fig. 3. The NPD construction results on four representative embedded network topologies, where colored triangles denote weighted sites. Top row: the input networks and site configurations, middle row: NPD results generated by direct assignment according to network power distance, and bottom row: results obtained by the proposed connected NPD construction method. From left to right: results on (a) tree-like branching graph, (b) ring-and-branch graph, (c) grid-like graph, and (d) bottleneck graph.

deformation stage, the capacities of network power cells are computed by Eq. (12) rather than by summing discrete demands. The capacity residuals are used to update site weights, and a connected NPD is then reconstructed using the same method as in the discrete MCCNPD.

The frozen-weight local refinement stage is modified only in terms of the repair unit. In the discrete case, a local move transfers a demand or a connected demand block. In contrast, the repair unit in the continuous-density case is a boundary micro-segment. Since a micro-segment carries line-integral mass, it can be split by length during boundary refinement. Assuming that a sublength Δl of micro-segment c is transferred, the corresponding moved mass is computed as $\Delta m = \rho_c \cdot \Delta l$. This allows finer capacity adjustment than moving indivisible demands. A local adjustment is accepted when the donor and receiver cells remain connected and the receiver capacity does not exceed its capacity. The final output is therefore a connected network power partition with maximum-capacity constraints, where the assigned demand of each network power cell is evaluated from the continuous network density field.

5. Experiments

We evaluate the proposed MCCNPD method on synthetic embedded network graphs and a real urban road network. The experiments examine four aspects, including the connectivity of the NPD construction, the behavior of the two-stage MCCNPD optimization, the computational performance under different input sizes and capacity constraints, and comparisons with representative network-based partitioning methods. Demands are represented either as discrete points or as a continuous

density field defined on network edges. All experiments are implemented in MATLAB and executed on a Windows 11 workstation with an Intel Core i7-12700KF CPU at 3.6 GHz and 64 GB RAM.

Unless otherwise stated, all experiments use the same default parameter setting. These parameters are selected according to three principles, i.e., preserving the geometric resolution of the network, avoiding instance-specific tuning, and ensuring finite termination of the heuristic optimization. The initial site weights are set to zero, and each discrete demand sample has unit mass in all experiments. The default capacity scale is set to 1.10, which produces a constrained but generally feasible setting. The influence is further evaluated in the sensitivity analysis by varying the capacity scale. The numbers of sites and demand samples are also treated as experimental variables in the following experiments. For the continuous-density experiments, the micro-segment resolution is determined from the network scale. Specifically, the target micro-segment length is computed based on Eq. (11), where $h_{min} = 1.0$ and $N_{micro} = 3000$. The maximum number of local repair iterations is 500, and the capacity tolerance is set to 10^{-9} . This setting follows the network density representation used in weighted network Voronoi analysis and provides a fixed numerical resolution for line-integral capacity accounting. The kernel type and bandwidth are kept fixed across all continuous-density experiments. For the optimization, the maximum iteration numbers and stall thresholds are used as termination safeguards. The first stage performs at most 80 weight-update iterations and uses an adaptive weight step with a base value of 0.80. The local power-margin quantile used for weight scaling is set to 0.20. The first stage uses bounded weight updates and records the best incumbent partition according to the capacity-violation measure. The second stage accepts only connectedness-preserving local refinements

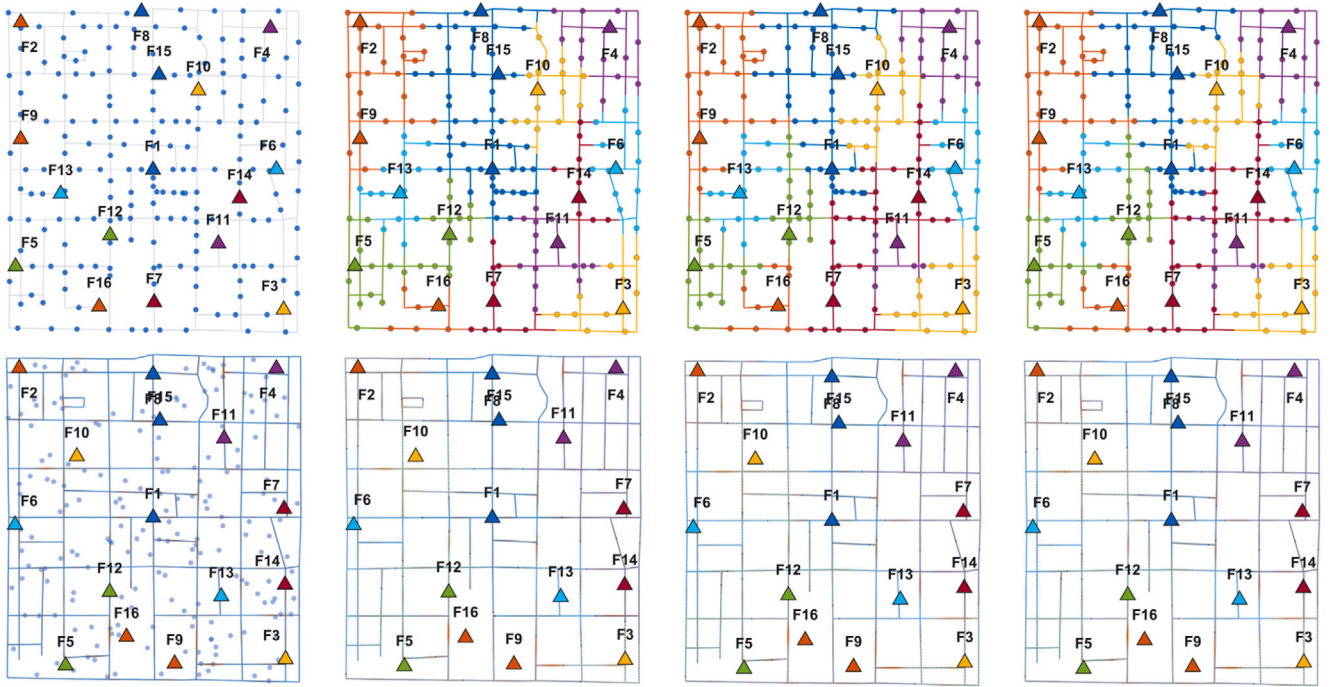


Fig. 4. MCCNPD optimization results under discrete and continuous demand settings. Top row: the optimization process for discrete demands, and bottom row: the corresponding process for a continuous demand-density field defined on the embedded network. From left to right: the input network with sites and demand distribution, the initial connected NPD, the result generated by the weight-driven global deformation stage, the final result obtained by the frozen-weight local refinement, and the convergence curves of weights and total overload.

that reduce capacity violation. These parameters allow the experiments to evaluate the robustness of the proposed method under different graph structures, site numbers, demand sizes, and capacity scales.

5.1. Connected NPD construction analysis

To examine the topological behavior of NPD construction, we evaluate the proposed connected NPD construction on four representative embedded graph configurations, including a tree-like branching graph, a ring-and-branch graph, a grid-like graph, and a bottleneck graph. These graphs cover typical network structures that may affect weighted shortest-path partitions, including branching terminals, cyclic routes, local shortcuts, and narrow separators. This experiment focuses on NPD construction, and demand or capacity constraints are not considered. For each graph, the same site configuration and weights are used for both constructions. The baseline directly assigns each graph element to the site with the minimum network power distance. The proposed method constructs cells through rooted graph expansion and explicitly enforces cell connectivity. The experimental results on these representative embedding graphs are presented in Fig. 3, where circles highlight the disconnected cells in the baseline method and are compared with the connected cells in our method.

The results in Fig. 3 demonstrate that direct minimization of network power distance cannot necessarily produce connected cells on embedded graphs. Although the direct assignment follows the point-wise lower envelope of weighted geodesic distance functions, a cell may reappear in spatially separated graph components. The highlighted regions show typical disconnected fragments on different graph structures, including tree branches, ring-induced discontinuities, shortcut-driven grid fragments, and unstable ownership near bottleneck transitions. In contrast, the proposed construction preserves rooted connectivity while still producing a complete partition of the graph. These results indicate that connectivity should be treated as an explicit geometric-topological constraint in the NPD construction.

5.2. MCCNPD results analysis

We further evaluate the proposed MCCNPD optimization under discrete demands and a continuous demand-density field defined on network edges. The same embedded graph and site configuration are used in both settings. In the discrete setting, the load of each network power cell is computed by summing the assigned demand samples. In the continuous setting, the demand points are projected onto the network to construct an edge-based density field, and the cell load is computed by integrating this density over the corresponding network segments. The capacity scale is set to the default value of 1.1. For each setting, we first construct an initial connected NPD without capacity constraints. The two-stage optimization is then applied. The first stage updates site weights to globally deform the network power cells. The second stage fixes the optimized weights and performs local boundary refinement to remove residual overload while preserving cell connectivity. The same default optimization parameters described above are used for both the discrete and continuous settings. Fig. 4 illustrates the overall optimization process of the proposed MCCNPD optimization method.

As shown in Fig. 4, we could observe that the initial NPD provides a valid connected partition, but several cells violate the maximum-capacity constraint. Through the first stage, the cell boundaries are redistributed over the graph, and the total overload decreases substantially. The visual results show that the optimization reshapes the network power distance field through site-weight adjustment. However, because ownership changes on embedded graphs are discrete and non-smooth, global weight deformation may leave residual overload near cell interfaces. The second stage resolves these local violations by transferring boundary demands or micro-segments under capacity and connectivity constraints. Consequently, the final partitions remain connected and satisfy the capacity constraints, demonstrating the effectiveness of the two-stage MCCNPD optimization.

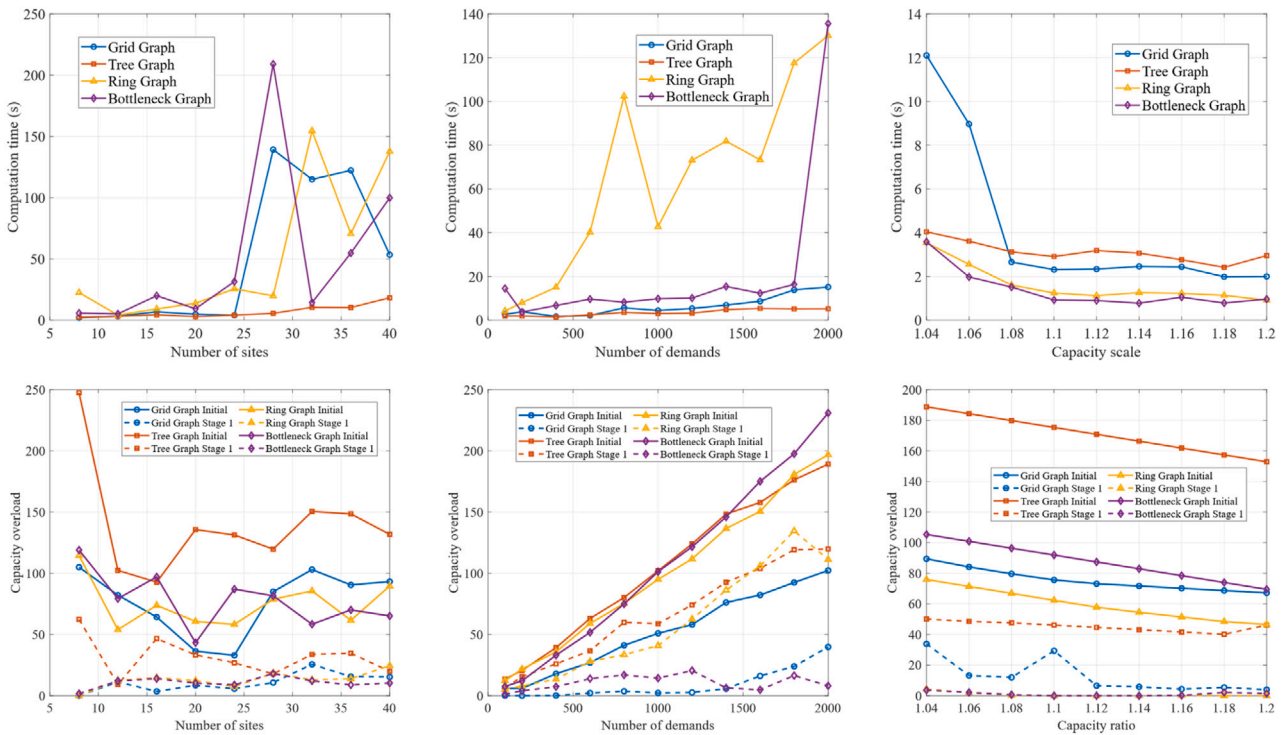


Fig. 5. Computational performance and capacity-violation analysis of the proposed MCCNPD optimization strategy. Top row: computation time with respect to the number of sites, the number of demands, and the capacity scale. Bottom row: total capacity overload of the initial connected NPD and the weight-driven global deformation results under the same three parameter settings. Notably, the frozen-weight local refinement reduces the final overload to zero within numerical tolerance in all reported cases.

Sensitivity and performance analysis **Sensitivity analysis.** We first evaluate the sensitivity of the proposed method with respect to the number of sites, the number of demands, and the capacity scale. Four embedded graph topologies are used in this experiment, and one factor is varied while the remaining parameters are fixed at their default values. The number of sites is varied from 8 to 40, and the number of demands is varied from 100 to 2000. The capacity scale is set from 1.04 to 1.20, and a larger capacity scale corresponds to a looser maximum-capacity constraint. The computation time in Fig. 5 measures the total cost of generating a connected NPD with maximum-capacity constraints, and the overload curves further verify the effectiveness of the two-stage optimization strategy.

The results in Fig. 5 show that the computational cost is jointly affected by input size, network topology, and capacity tightness. Increasing the number of sites or demand samples generally increases the running time because more cells, boundary candidates, and demand assignments must be processed. The computational cost curves are not strictly monotonic, especially on the ring-and-branch and bottleneck graphs. This behavior is consistent with graph-embedded power diagrams, where small changes in sites or demands may alter cell adjacency, boundary ownership, and the number of local refinement steps. Increasing the capacity scale reduces the running time in most cases because looser capacity constraints require fewer local refinements after the global weight deformation.

The overload curves in the bottom row of Fig. 5 further show the effectiveness and the empirical convergence behavior of the proposed heuristic two-stage optimization strategy. The initial connected NPD is directly constructed from the network power distance without capacity optimization, which may lead to substantial capacity violations. The total capacity violation after the first stage is consistently reduced across different site numbers, demand numbers, capacity scales, and graph topologies. The reduction is especially evident in cases with large initial overloads, demonstrating that the weight-driven deformation provides an effective global correction of cell masses. However, these capacity

violations cannot be fully removed because demand assignments on an embedded graph change discretely and the remaining overload may be localized near cell boundaries. The second stage subsequently applies frozen-weight local refinement to remove these residual violations. The final overload after the second stage in all reported settings is reduced to zero within the numerical tolerance. These results indicate that the proposed two-stage heuristic terminates stably and effectively reduces capacity violations under the tested graph structures and parameter settings.

Stage-wise computational performance. To further analyze the computational behavior of the proposed method, we conduct an additional stage-wise performance experiment on four controlled graph scales. The four embedded network graphs, denoted GS-1, GS-2, GS-3, and GS-4, are generated from grid-like urban network layouts with increasing graph sizes. They contain 81 vertices and 146 edges (GS-1), 169 vertices and 314 edges (GS-2), 289 vertices and 546 edges (GS-3), and 441 vertices and 842 edges (GS-4), respectively. Two groups of experiments are performed. In the first group, the number of sites varies from 10 to 40, while the number of demands is fixed at 1000. In the second group, the number of demands is varied from 500 to 2000, while the number of sites is fixed at 20. The capacity scale is set to 1.10 in all tests. For each experiment, we separately record the time for one connected NPD construction, Stage 1 weight optimization, stage 2 local refinement, and the total computation time.

The computational performance of these results is summarized in Fig. 6. The connected NPD construction includes graph construction, site and demand embedding, rooted network power expansion, and boundary splitting. The time for a single connected NPD construction grows mildly with graph scale and demand size. The computation time of the weight-driven global deformation (Stage 1) demonstrates a clear dependence on graph and demand scales since it repeatedly reconstructs connected NPDs during the weight-update iterations. The computation time of the weight-frozen local refinement (Stage 2) is mainly determined by the difficulty of removing the residual capacity

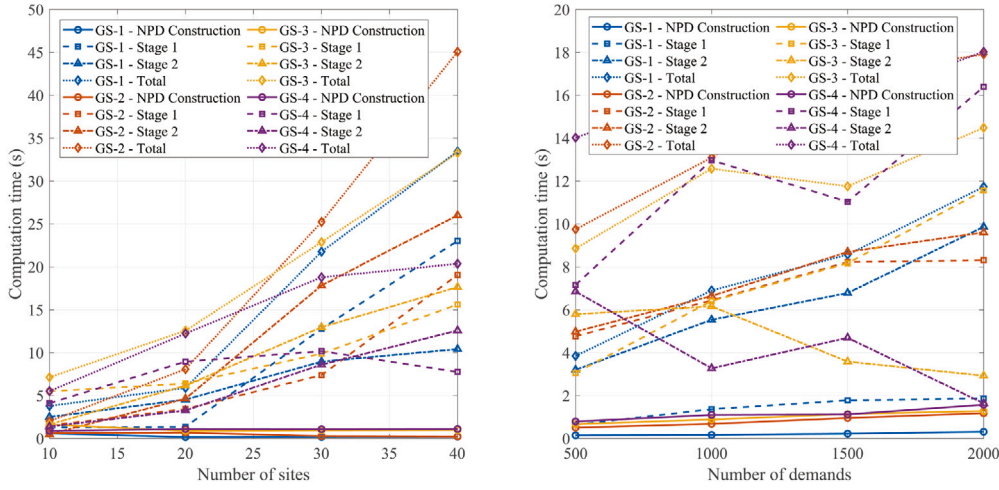


Fig. 6. The stage-wise computational performance of the proposed MCCNPD optimization strategy. Left: computation time with respect to the number of sites in various graph scales, and right: computation time with respect to the number of demands in various graph scales.

Table 1

Computational performance and final overload ratio of different partitioning methods on representative network graphs, corresponding to the results in Fig. 7. The overload ratio is defined as the final total overload divided by the total demand.

Methods	Sites	Demands	Computation time (s)				Final overload ratio			
			Tree	Ring	Grid	Bottleneck	Tree	Ring	Grid	Bottleneck
WNVD	10	500	0.015	0.039	0.175	0.029	37.78%	26.01%	16.70%	23.59%
	20	1000	0.113	0.030	0.075	0.018	23.61%	17.98%	18.75%	20.24%
CCNVD	10	500	8.106	9.865	8.742	10.492	8.82%	0.00%	0.00%	0.00%
	20	1000	45.745	26.084	38.557	14.397	12.94%	0.11%	0.02%	0.07%
Ours-D	10	500	10.980	10.252	7.199	4.697	0.00%	0.00%	0.00%	0.00%
	20	1000	24.360	30.780	16.863	20.714	0.00%	0.00%	0.00%	0.00%
Ours-C	10	500	11.646	14.958	13.546	12.041	0.00%	0.00%	0.00%	0.00%
	20	1000	14.290	22.016	25.014	28.006	0.00%	0.00%	0.00%	0.00%

violations. In particular, large site counts may create more boundary conflicts and require more local transfers, leading to more time consumption for local refinement. In contrast, a nearly feasible partition produced by the weight-driven global deformation stage may greatly reduce the computation time for the frozen-weight local refinement stage.

5.3. Comparison

We compare the proposed method with representative network-based partitioning approaches on four embedded graph topologies. The WNVD [23] is used as a distance-based baseline without capacity constraints. The CCNVD [11] is used as a capacity-aware baseline based on nearest-network-distance partitions. The proposed method is evaluated under both discrete demands and continuous network-density fields, denoted as Ours-D and Ours-C, respectively. For each topology, two experimental settings are tested, including 10 sites with 500 demands and 20 sites with 1000 demands. The capacity scale is set to the default value of 1.1 for all capacity-constrained methods. The same site and demand configurations are used across methods whenever applicable. The qualitative results are presented in Fig. 7, and the related computational performance is reported in Table 1. Note that we report the total capacity overload as the sum of all positive capacity violations and define the overload ratio as the total overload divided by the total demand. A value of zero indicates that all cells satisfy their maximum-capacity constraints within the numerical tolerance.

The visual comparison in Fig. 7 shows clear differences among these methods. The WNVD produces network-based weighted partitions efficiently, but it is distance-driven and lacks capacity constraints. This is also reflected in Table 1, where the WNVD produces large final

overload ratios in all tested cases. The CCNVD generates capacity-aware contiguous regions and reduces overload substantially compared with the WNVD. However, the partition structure is still governed by the nearest network distance and several cases retain small capacity violations. In contrast, the proposed Ours-D and Ours-C methods generate connected network power cells under maximum-capacity constraints. Their final overload ratios are zero in all reported cases, indicating that the two-stage MCCNPD optimization successfully removes the residual violations left by the initial connected NPD and weight-driven global deformation.

The computation times in Table 1 reflect the trade-off between partition complexity and constraint satisfaction. The WNVD is the fastest method because this method performs a one-shot weighted network assignment without capacity optimization. The CCNVD and the proposed methods require additional optimization to satisfy capacity constraints, leading to higher computational costs. The proposed discrete method is generally comparable to the CCNVD, while the continuous-density version introduces extra costs from line-integral density evaluation and micro-segment refinement. Overall, the comparison results illustrate that the proposed method provides stronger capacity feasibility than distance-only weighted network partitions while preserving the structure of network power cells.

Ablation analysis To evaluate the contribution of the main components in the proposed method, we conduct an ablation experiment on a ring-branch network graph. This graph contains circular, radial, and terminal-branch structures, which make both topological consistency and capacity adjustment nontrivial. We use the same eight sites, 260 discrete demand points, a capacity scale of 1.08, and identical initial power weights for all configurations. Four variants are compared in this experiment, i.e., direct network power assignment without the connected construction, the proposed pipeline without the weight-driven

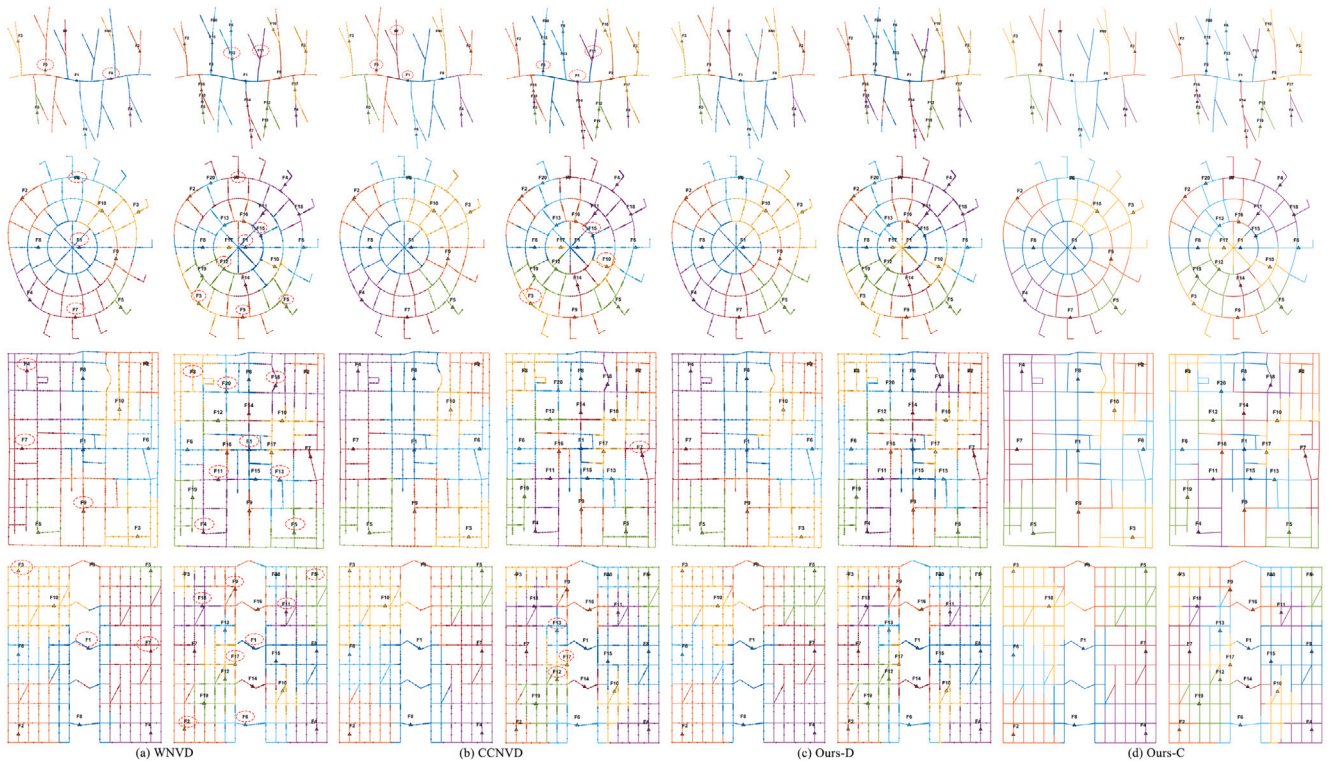


Fig. 7. Qualitative comparison of different partitioning methods on four representative network graphs. For each network graph, two experimental settings are considered, including one with 10 sites and 500 demands (left) and another with 20 sites and 1000 demands (right). From top to bottom: results on tree, ring, grid, and bottleneck graphs. Notably, Ours-D refers to the proposed method with discrete demands, whereas Ours-C denotes the proposed method with continuous-density demands.

Table 2
Quantitative ablation results of the proposed MCCNPD method, corresponding to these visual results in Fig. 8.

Results	Connected	Violating cells	Total violation	Capacity feasible
Fig. 8(a)	No	3	29.70	No
Fig. 8(b)	Yes	1	0.90	No
Fig. 8(c)	Yes	2	2.80	No
Fig. 8(d)	Yes	0	0	Yes

global deformation, the proposed pipeline without the frozen-weight local refinement, and the full method. These ablation experimental results are summarized in Fig. 8 and Table 2.

According to the results in Fig. 8 and Table 2, it could be observed that without the connected NPD construction, direct network power assignment produces one disconnected cell and three capacity-violating cells. This confirms that the network-power distance field alone cannot guarantee topologically valid cells on embedded networks. When the first stage is removed, the partition remains connected, but the second stage alone leaves one residual capacity-violating cell. When the second stage is removed, the first stage reduces the capacity imbalance through weight-driven global deformation, but two cells still violate the maximum-capacity constraints. In contrast, the full method produces connected cells with zero capacity-violating cells and zero total capacity violation. These results demonstrate that the three main components play complementary roles in the proposed MCCNPD method.

6. Real-city application

To further evaluate the applicability of the proposed method to a real urban road network, we conduct an application study on the main urban road network of Nanjing, China. The road network is

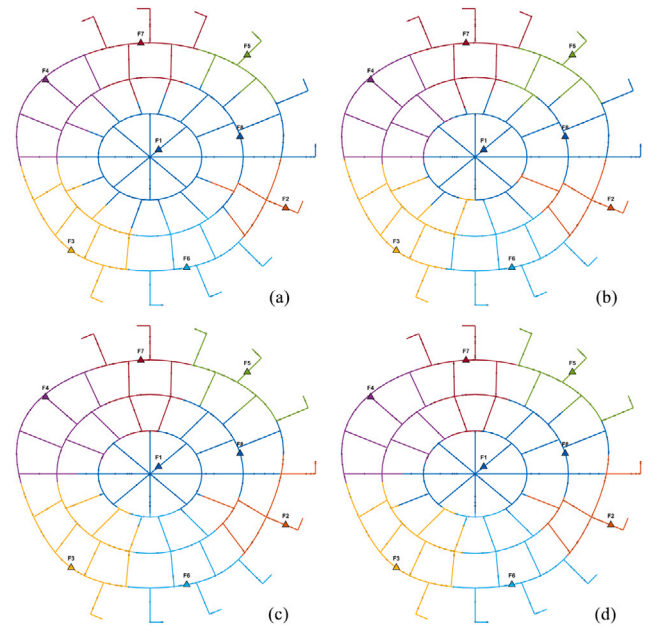


Fig. 8. Ablation experimental results of the proposed MCCNPD method on a ring-branch network graph. (a) Direct network power assignment without the connected construction produces a disconnected cell and capacity violations. (b) Removing the weight-driven global deformation stage preserves connectedness but leaves a residual capacity violation after local refinement. (c) Removing the frozen-weight local refinement stage leaves overloaded cells after weight-driven global deformation. (d) The full method produces connected and capacity-feasible network power cells.

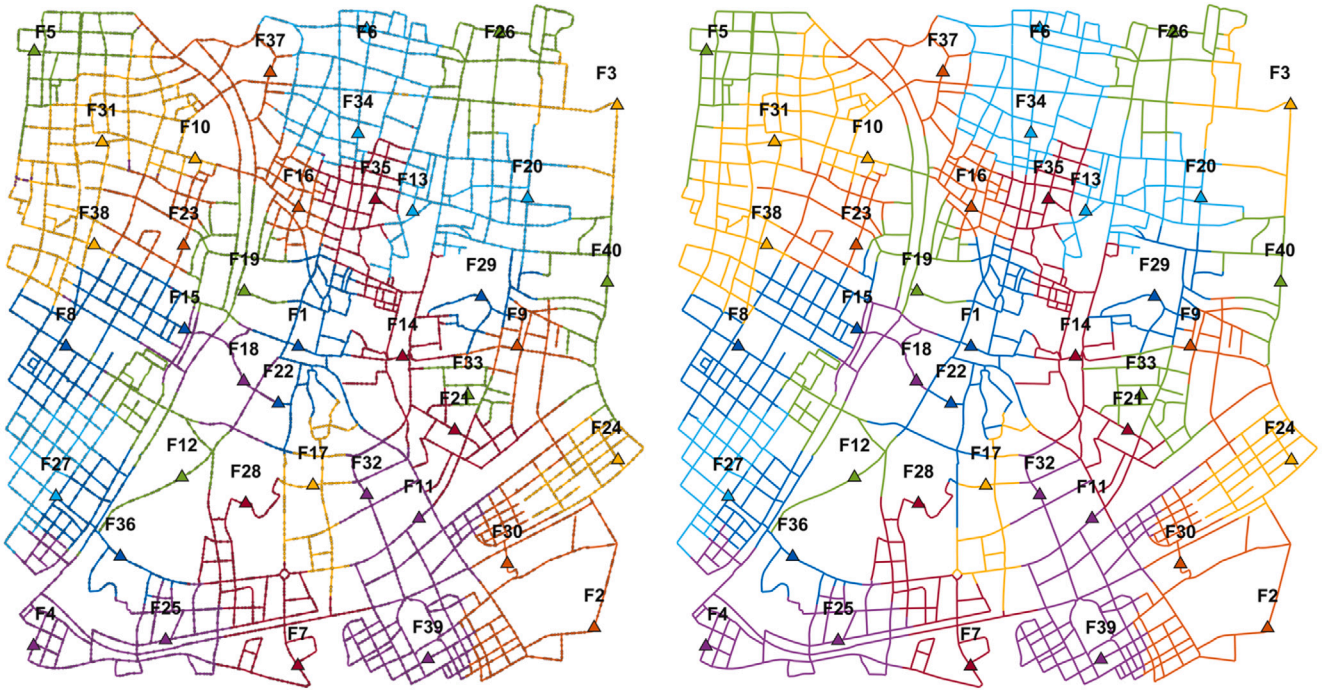


Fig. 9. Embedded network partitions based on the proposed MCCNPD on the main urban road network of Nanjing under the discrete demand (left) and the continuous-density (right) settings.

extracted from OpenStreetMap and converted into an embedded graph after preprocessing. The resulting graph contains 5551 vertices and 6157 segments, with a total road length of 486.60 km. Notably, extremely narrow roads are excluded, and thoroughfares of the main urban road network are primarily concentrated in our statistics. We randomly sample 40 sites and 10,000 demands with random values on the network. The capacity scale is set to 1.1, which is the same as the default setting in the synthetic experiments. In the discrete setting, the 10,000 demand samples are directly used to compute the load of each network power cell. In the continuous-density setting, the same samples are projected onto the road network to construct an edge-based demand-density field. Network power cell loads are then evaluated by integrating the density along network segments. The experimental results of the proposed method on the urban road network of Nanjing are provided in Fig. 9, and the quantitative statistics of the generated results are provided in Table 3.

The visual results in Fig. 9 demonstrate that the proposed method can generate connected network power cells with capacity constraints on a large and irregular urban road network. The Nanjing network contains dense local streets, arterial corridors, irregular blocks, and non-uniform connectivity, which make region generation more challenging than on synthetic graphs. The discrete and continuous settings produce different cell boundary patterns because they use different demand representations. The discrete result reflects the spatial distribution of sampled demands, while the continuous-density result produces smoother mass redistribution along road segments. The quantitative statistics in Table 3 further support the capacity feasibility of the final road-network partitions. The assigned demands of each network power cells remains below or equal to the maximum-capacity constraint within numerical tolerance. The optimized weights vary substantially across sites, indicating that the proposed method deforms the network power distance field to compensate for heterogeneous network structure and demand distribution. Compared with the discrete setting, the continuous-density setting generally produces smoother mass values because boundary refinement could transfer fractional line-integral mass rather than indivisible demands. These results demonstrate that

the proposed MCCNPD framework could be applied to real urban road-network region generation while preserving connected cells and satisfying maximum-capacity constraints.

7. Conclusion

This paper introduces network power diagrams with maximum-capacity constraints for graph-embedded network partitioning. By replacing Euclidean power distance with shortest-path-based network power distance, we extend power diagram construction to embedded network graphs. To generate connected network power diagrams, we propose a connected-by-construction NPD method based on rooted graph growth. Based on this, a two-stage optimization strategy is proposed to satisfy maximum-capacity constraints, combining weight-driven global deformation with frozen-weight local refinement. The formulation is further extended from discrete demands to continuous network demand densities through line-integral load evaluation. Experiments on synthetic networks and a real urban road-network demonstrate that the proposed method can generate connected and capacity-constrained network power cells under different topologies, demand models.

Limitations and future work. However, there are still several limitations in our work. The proposed optimization framework is heuristic and lacks a complete theoretical proof of global convergence or global optimality. Despite the algorithm has finite termination under the implemented update and refinement rules, it may stop with residual capacity violation when no admissible connectedness-preserving improving transfer exists. This may occur under extremely tight capacity constraints, severe bottleneck structures, or highly fragmented demand distributions. In addition, the continuous-density extension relies on finite micro-segment discretization and therefore provides a numerical approximation rather than an analytic continuous solution. In the future work, we will investigate severe capacity shortage, stronger feasibility conditions, adaptive density discretization, and more global search strategies for difficult network configurations.

Table 3

Computational results of capacity upper bounds, assigned demands, and optimized site weights in the Nanjing real-city case study under both discrete-demand and continuous-density settings, where c_i^{max} denotes the maximum-capacity of each network power cell, m_i represents the final assigned cell demands, and w_i provides the optimized site weights.

Facility	Discrete			Continuous		
	c_i^{max}	m_i	w_i	c_i^{max}	m_i	w_i
1	206.71	206.37	131.49	206.71	170.55	135.87
2	206.71	205.97	132.96	206.71	192.91	93.32
3	206.71	85.06	193.81	206.71	168.98	209.35
4	206.71	206.28	230.04	206.71	197.03	139.63
5	206.71	205.79	-424.30	206.71	206.71	-207.95
6	206.71	206.22	60.16	206.71	199.95	-10.93
7	206.71	206.37	120.00	206.71	172.36	5.42
8	206.71	206.63	-399.53	206.71	200.93	-338.41
9	206.71	199.58	115.14	206.71	187.45	131.80
10	206.71	206.45	-221.42	206.71	206.71	-139.82
11	206.71	201.80	-13.18	206.71	186.16	22.48
12	206.71	206.21	185.30	206.71	204.59	163.87
13	206.71	206.63	51.77	206.71	182.53	-72.22
14	206.71	206.45	123.61	206.71	186.20	81.09
15	206.71	206.58	-19.66	206.71	205.98	45.44
16	206.71	206.35	-33.08	206.71	194.66	-92.18
17	206.71	151.21	119.68	206.71	168.82	107.01
18	206.71	180.18	167.46	206.71	145.03	228.44
19	206.71	206.57	91.08	206.71	188.65	104.85
20	206.71	206.70	45.74	206.71	179.51	-79.21
21	206.71	192.64	118.69	206.71	172.14	178.51
22	206.71	57.97	119.39	206.71	202.25	240.11
23	206.71	206.33	-113.24	206.71	206.71	-42.35
24	206.71	190.54	-132.80	206.71	187.00	-64.36
25	206.71	206.62	-1.21	206.71	187.99	-29.22
26	206.71	206.65	54.25	206.71	184.64	-47.53
27	206.71	206.55	-242.32	206.71	205.66	-282.02
28	206.71	156.97	200.53	206.71	172.35	142.90
29	206.71	90.77	129.17	206.71	164.07	172.06
30	206.71	206.69	-114.19	206.71	185.08	-90.86
31	206.71	206.70	-339.76	206.71	206.71	-226.90
32	206.71	205.47	91.92	206.71	174.18	87.68
33	206.71	69.49	106.30	206.71	166.13	219.45
34	206.71	206.52	19.23	206.71	198.25	-68.23
35	206.71	206.34	57.03	206.71	189.12	-56.97
36	206.71	206.68	-0.14	206.71	206.71	-42.81
37	206.71	206.69	-185.13	206.71	206.71	-208.12
38	206.71	206.33	-318.60	206.71	206.71	-221.09
39	206.71	206.23	-273.97	206.71	197.08	-276.17
40	206.71	161.34	167.77	206.71	151.66	88.07

CRedit authorship contribution statement

Yuyou Yao: Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Shouyan Xia:** Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Ensheng Liu:** Writing – review & editing, Writing – original draft, Software, Resources, Investigation, Conceptualization. **Chengjie Gu:** Software, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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